

Supplementary Material*

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* This supplementary material was provided by the authors to give readers further details on their article. The material was reviewed but not copyedited.

Supplement Table 1. Breast Cancer-Specific Risk Differences and Hazard Ratios for Continuing vs. Stopping Screening by 5-Year Age Groups in the Emulated Target Trials, Medicare 1999-2008. Number in parentheses are 95% confidence intervals

	Age 70-74		Age 75-79		Age 80-84	
	Continue	Stop	Continue	Stop	Continue	Stop
Women	1,235,459	1,235,459	923,604	923,604	480,131	480,131
Person-years	3,518,476	2,067,098	2,482,653	1,647,934	1,169,013	941,433
Breast cancer deaths	601	479	563	477	369	348
8-year breast cancer mortality difference, %	-1.0 (-2.3 to 0.1)	Ref.	-0.4 (-1.2 to 1.2)	Ref.	0.7 (-1.2 to 3.0)	Ref.
Hazard ratio	0.78 (0.63 to 0.95)	Ref.	0.86 (0.69 to 1.12)	Ref.	1.17 (0.85 to 1.54)	Ref.

Supplement Table 2. Point Estimates for the 8-Year Breast Cancer Death Risk Differences With Different Levels of Adjustment, Medicare 1999-2008.

Level of adjustment	8-year breast cancer death risk difference, cases per 1000 women	
	Age group 70-74	Age group 75-84
None	-5.6	-2.5
Baseline variables ¹	-4.9	-1.5
Baseline variables ¹ and time-varying screening history ²	-2.6	-1.2
Baseline variables ¹ , time-varying screening history ² and time-varying use of health care resources ³	-2.4	-1.3
Full adjustment for baseline variables ¹ , time-varying screening history ² , time-varying use of health care resources ³ and time-varying comorbidities ⁴	-1.0	0.07

¹Baseline variables: age, calendar year, race, census region, visits to the ER in the 6 months before baseline, days admitted in the 6 months before baseline, combined comorbidity score at baseline; reception of the following in the year before baseline: influenza vaccine, bone mass measurement, CVD screening, diabetes screening, annual wellness visit, colorectal cancer screening, pelvic screening; diagnosis of the following before baseline: Alzheimer's disease and related disorders, acute myocardial infarction in the previous year, atrial fibrillation, anemia, asthma, cataracts, chronic heart failure, chronic kidney disease, COPD, colorectal cancer, depression, diabetes, endometrial cancer, glaucoma, hip fracture in the previous year, hypertension, hypothyroidism, ischemic heart disease, hyperlipidemia, lung cancer, osteoporosis, rheumatoid arthritis/osteoarthritis, stroke in the previous year

²Screening history variables: time since last mammogram, type (screening, diagnostic) and number of mammograms received, breast symptoms in the previous six months.

³Health care resource use variables: visits to the ER in the previous 6 months, days admitted in the previous 6 months, reception of the following in the last year: influenza vaccine, bone mass measurement, CVD screening, diabetes screening, annual wellness visit, colorectal cancer screening, pelvic screening.

⁴Comorbidities variables: combined comorbidity score, Alzheimer's disease and related disorders, acute myocardial infarction in the previous year, atrial fibrillation, anemia, asthma, cataracts, chronic heart failure, chronic kidney disease, COPD, colorectal cancer, depression, diabetes, endometrial cancer, glaucoma, hip fracture in the previous year, hypertension, hypothyroidism, ischemic heart disease, hyperlipidemia, lung cancer, osteoporosis, rheumatoid arthritis/osteoarthritis, stroke in the previous year, long term care center resident.

Supplement Table 3. Deterministic Sensitivity Analysis for Unmeasured Confounding. True 8-year risk difference of breast cancer mortality for continue vs. stop screening under different scenarios with a hypothetical, dichotomous, time-fixed confounder that has not been measured and that is unassociated with all the measured confounders. The estimated 8-year risk differences of breast cancer mortality were -1.00 per 1000 women in the age group of 70-74 and 0.07 per 1000 women in the age group of 75-84

Age group	Prevalence of the unmeasured confounder		8-year risk difference of breast cancer mortality*	Corrected 8-year risk difference of breast cancer mortality
	“continue screening”	“stop screening”	(presence vs. absence of the confounder, cases per 1000 women)	(continue vs. stop screening, cases per 1000 women)
70-74	1%	5%	-0.5	-1.02
	1%	5%	0.5	-0.98
	10%	5%	-0.5	-0.975
	10%	5%	0.5	-1.025
75-79	1%	5%	-0.5	0.05
	1%	5%	0.5	0.09
	10%	5%	-0.5	0.095
	10%	5%	0.5	0.045

*Assumed to be constant conditional on treatment group.

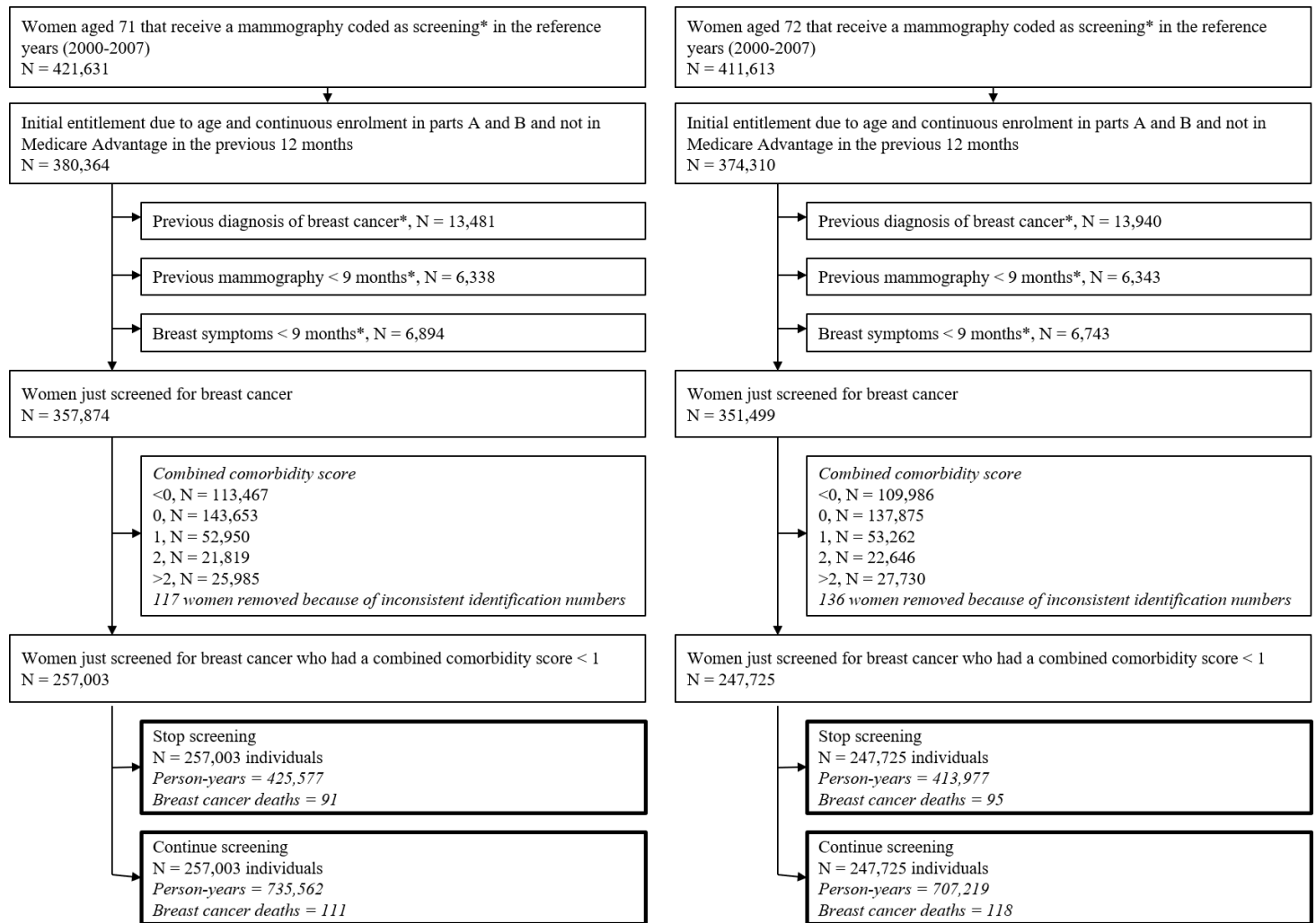
Supplement Table 4. Probabilistic Sensitivity Analysis for Unmeasured Confounding. True 8-year risk difference of breast cancer mortality for continue vs. stop screening under different scenarios with a hypothetical, dichotomous, time-fixed confounder that has not been measured and that is unassociated with all the measured confounders. The estimated 8-year risk differences of breast cancer mortality were -1.00 per 1000 women (95% CI: -2.3 to 0.1) in the age group of 70-74 and 0.07 per 1000 women (95% CI: -0.93 to 1.3) in the age group of 75-84.

Age group	Min	Percentile 2.5	Percentile 25	Median	Percentile 75	Percentile 97.5	Max
70-74	-1.16	-1.06	-1.01	-0.99	-0.97	-0.90	-0.79
75-84	-0.09	0.01	0.06	0.08	0.10	0.17	0.28

Supplement Table 5. 8-Year Cancer-Specific Death Risk Difference, Cases per 1000 Individuals and Hazard Ratio, Negative Control Outcome (Death Because of Cancer of Corpus Uteri). Number in parentheses are 95% confidence intervals

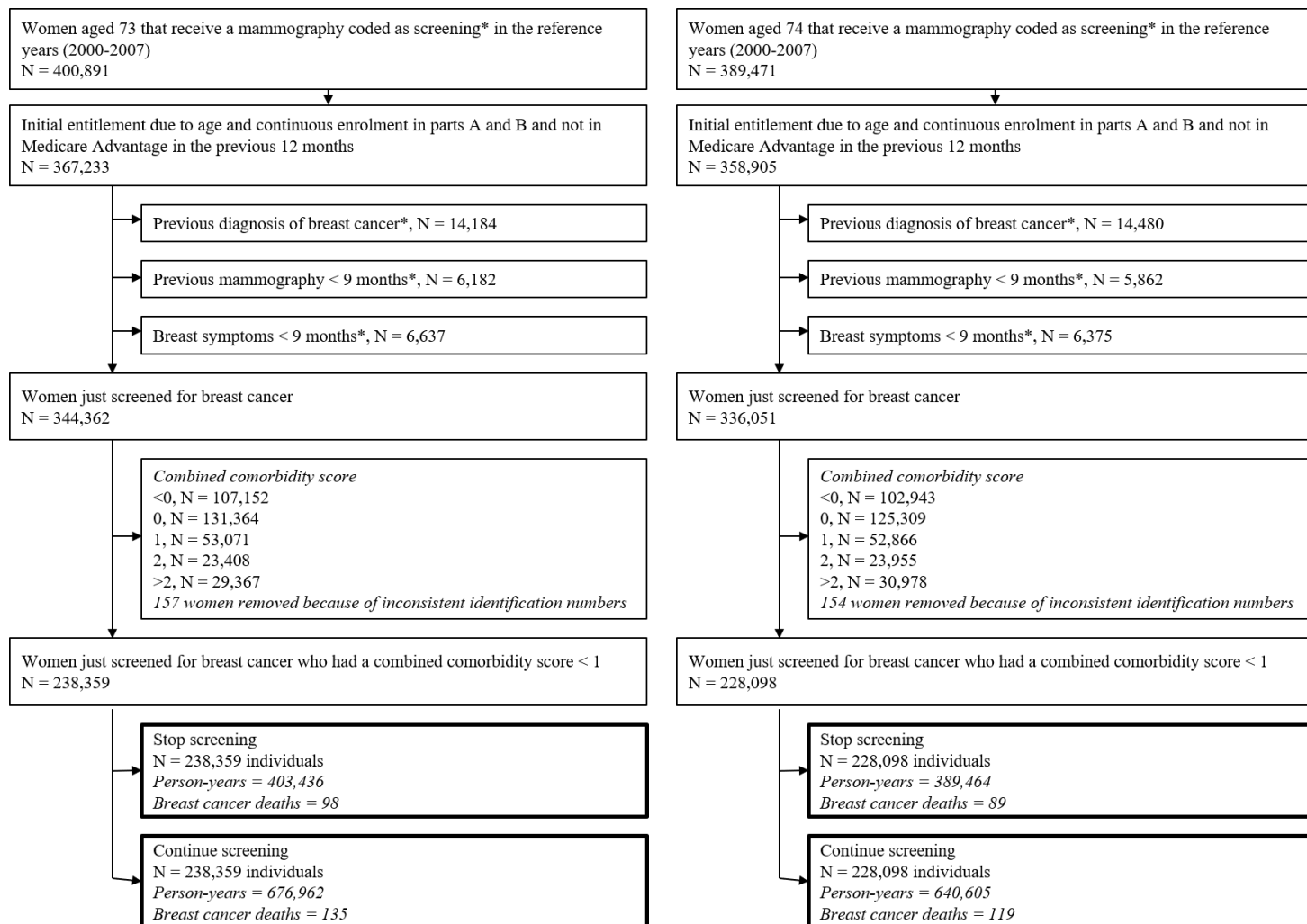
	Age 70-74		Age 75-84	
	Continue	Stop	Continue	Stop
Deaths	204	160	316	231
HR	0.83 (0.50 to 1.07)	Ref.	1.11 (0.81 to 1.36)	Ref.
8-year cancer mortality difference, ‰	0.02 (-0.28 to 0.23)	Ref.	0.10 (-0.21 to 0.44)	Ref.

Supplement Figure 1. Selection of Eligible Patients Aged 71-84, Medicare 1999-2008.



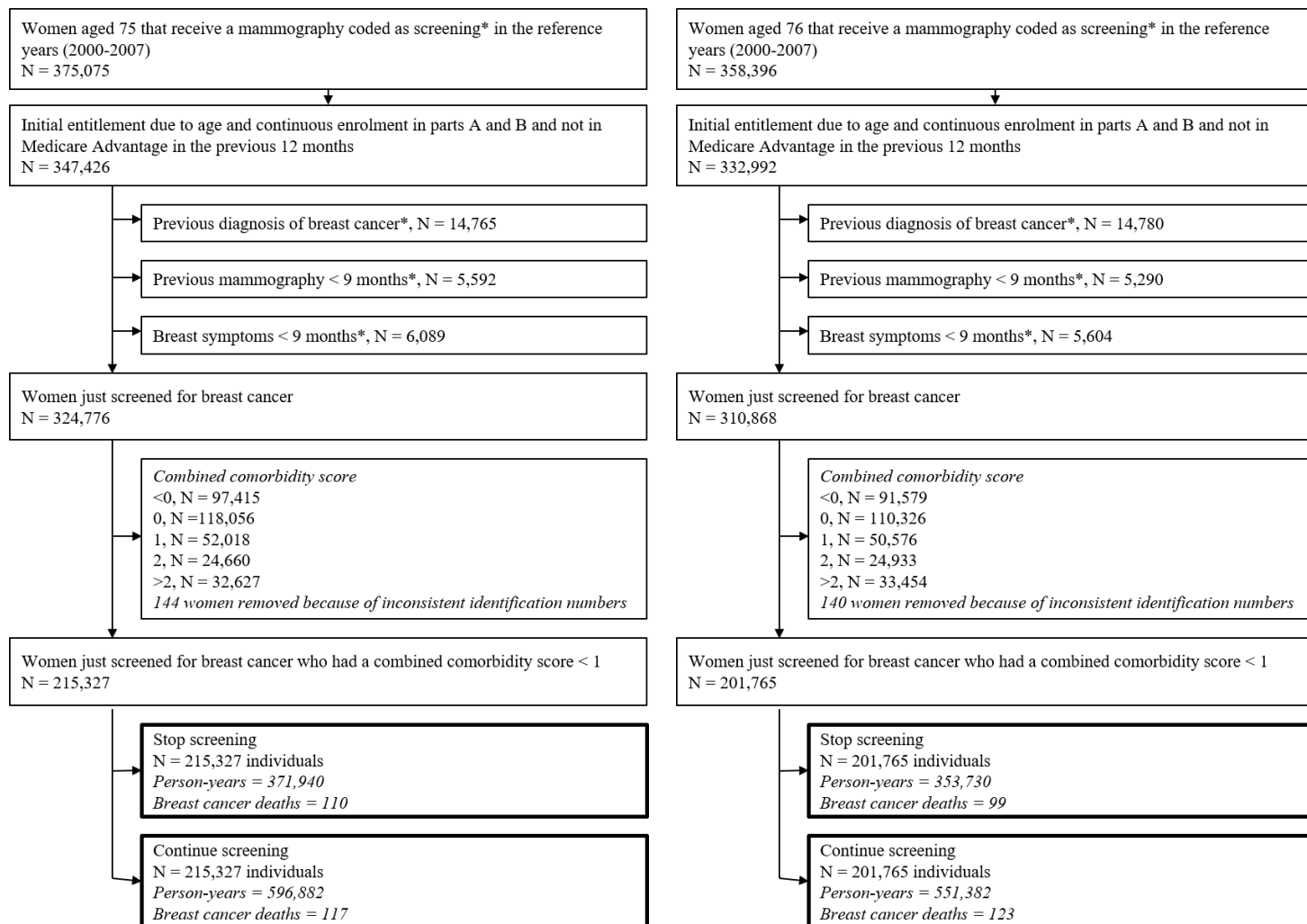
*Required by the screening mammography extraction algorithm.

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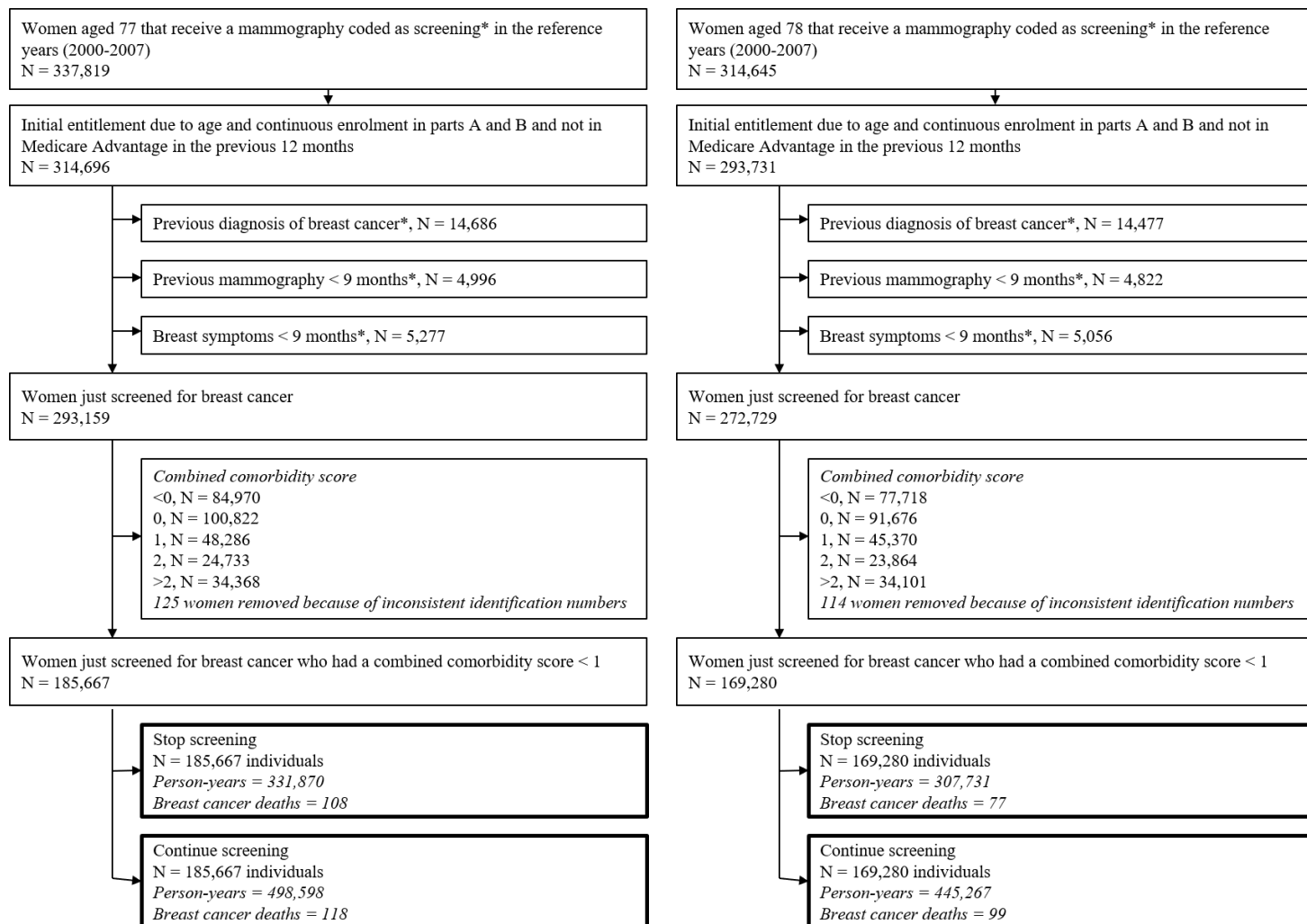
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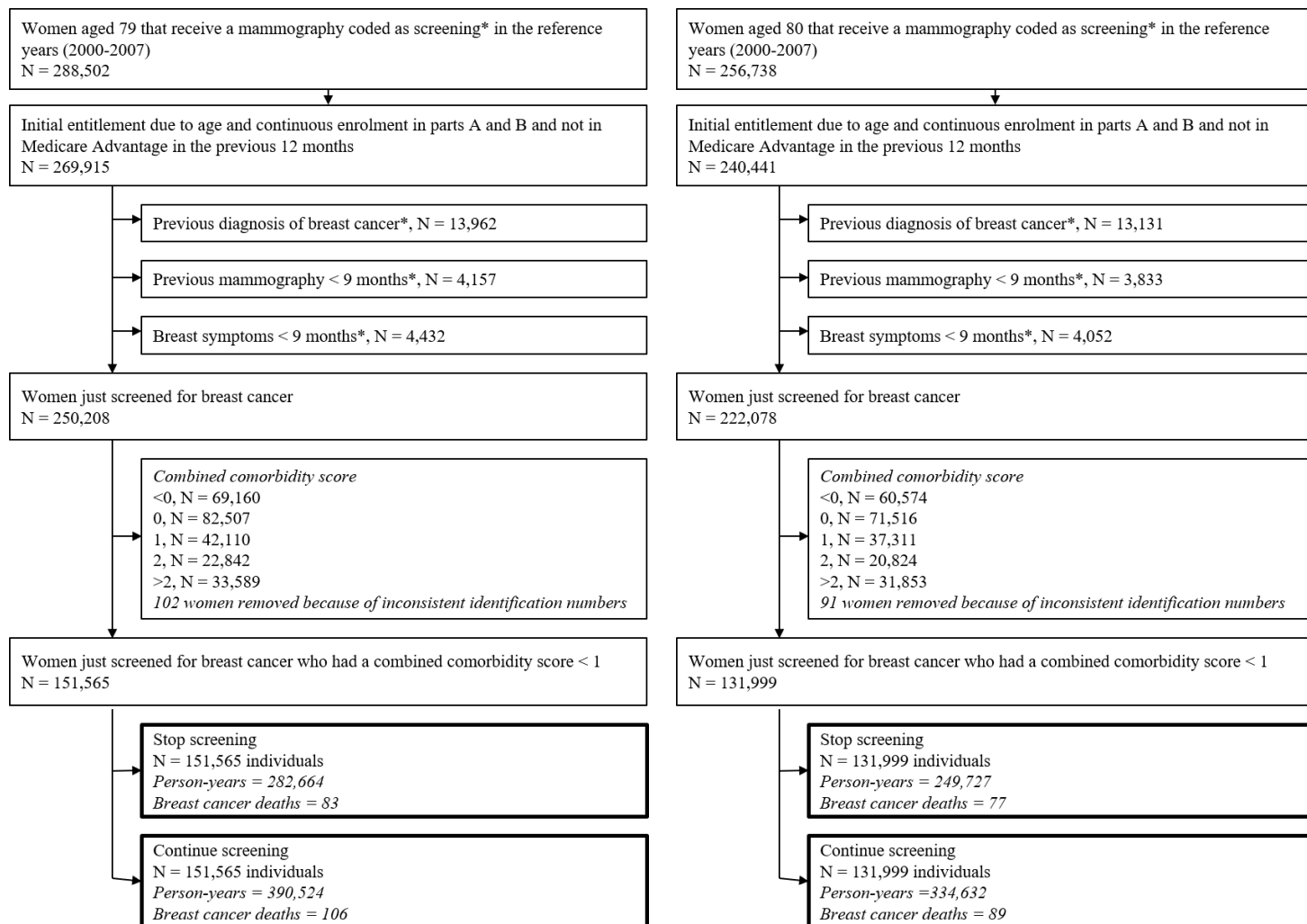
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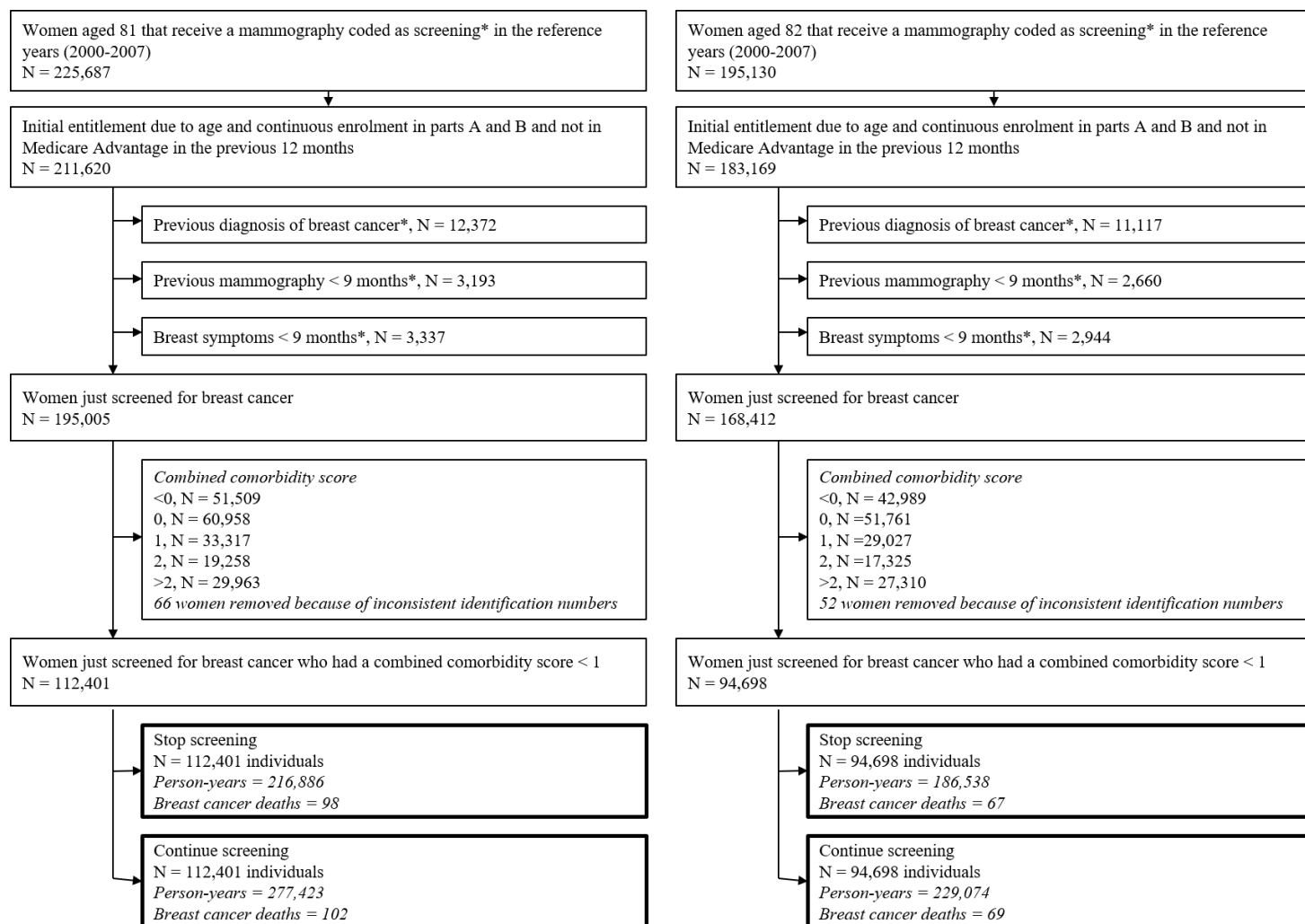
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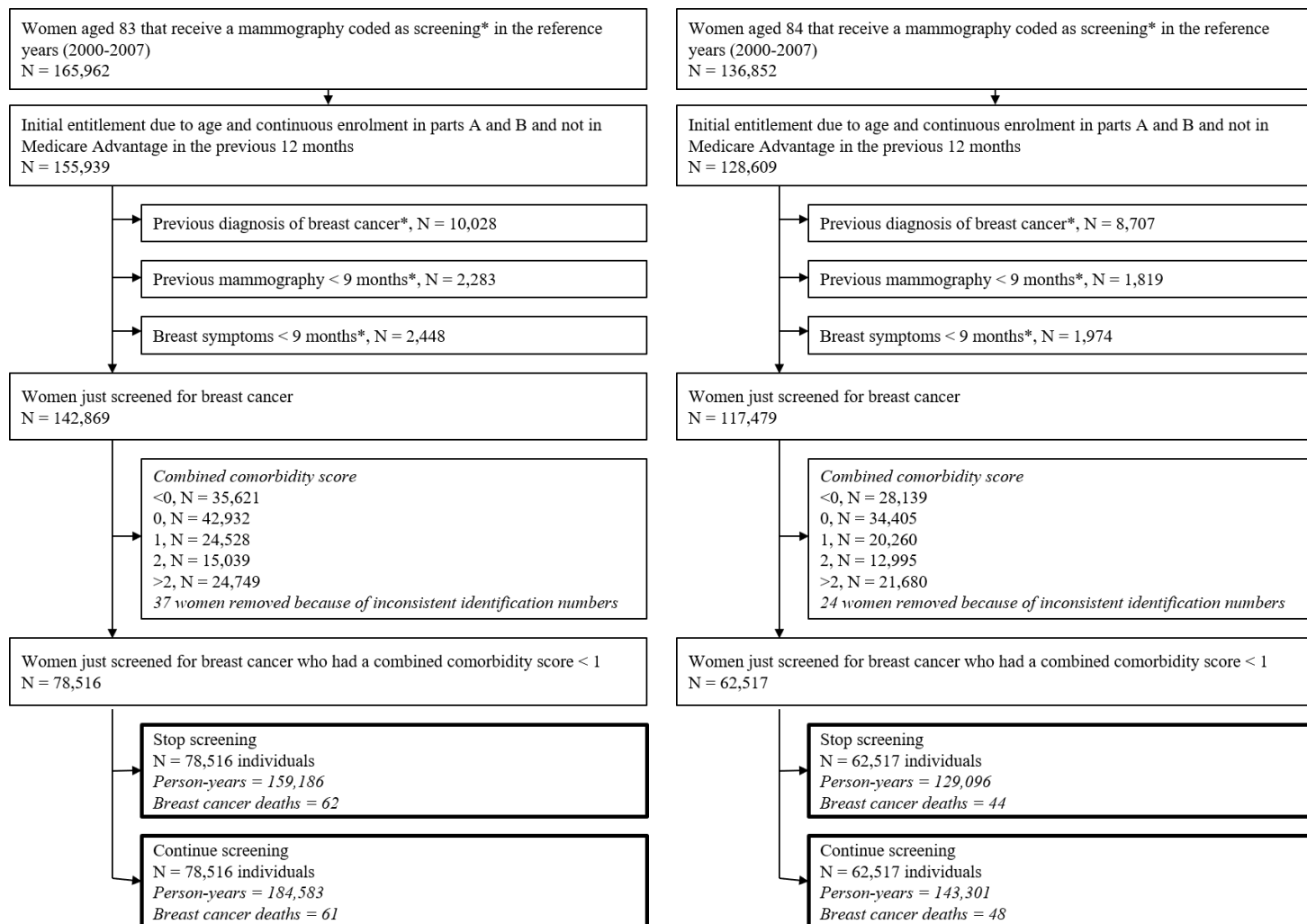
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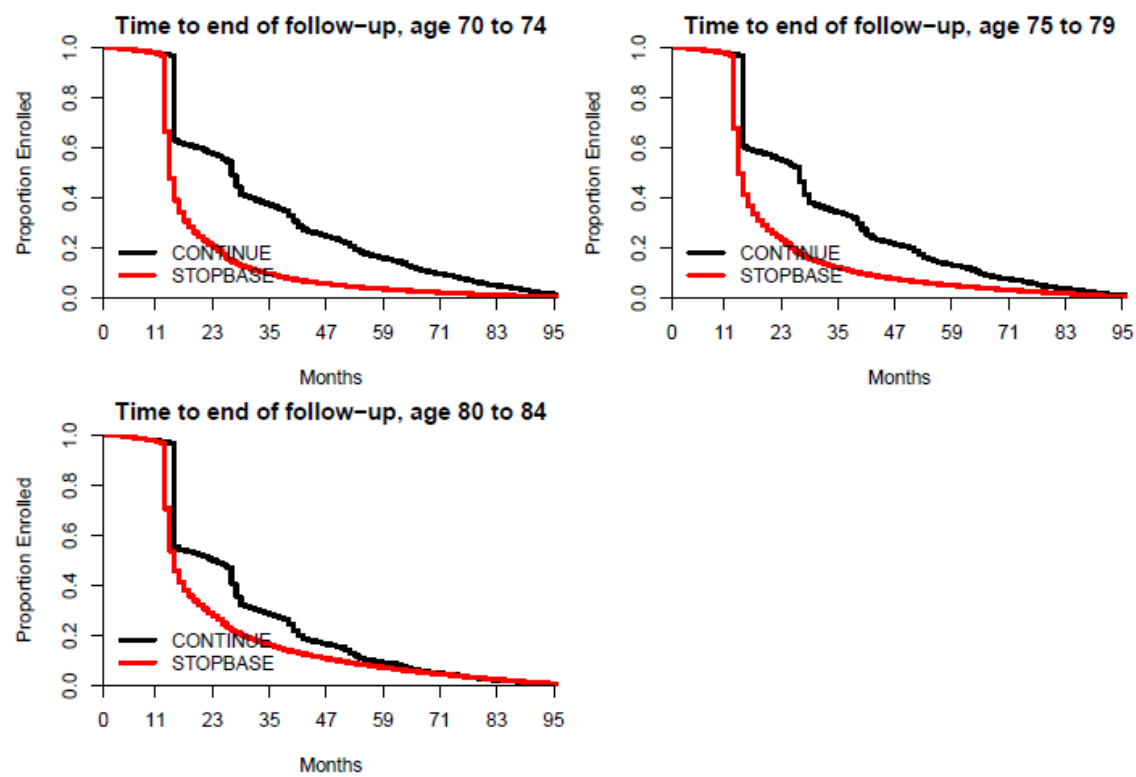
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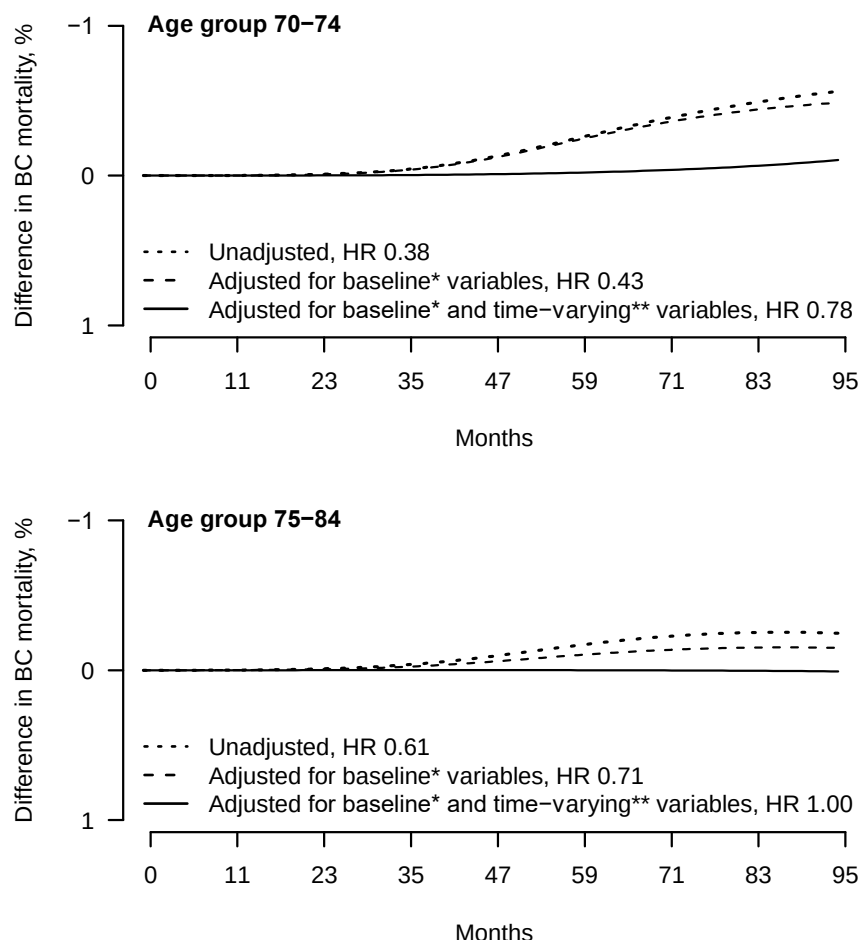
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Supplement Figure 2. Time to the End of Follow-up in Each Screening Strategy by 5-Year Age Group.



Supplement Figure 3. Effect of Different Levels of Adjustment on the Breast Cancer (BC)-Specific Mortality Difference for Continuing vs. Stopping Screening, Medicare 1999-2008



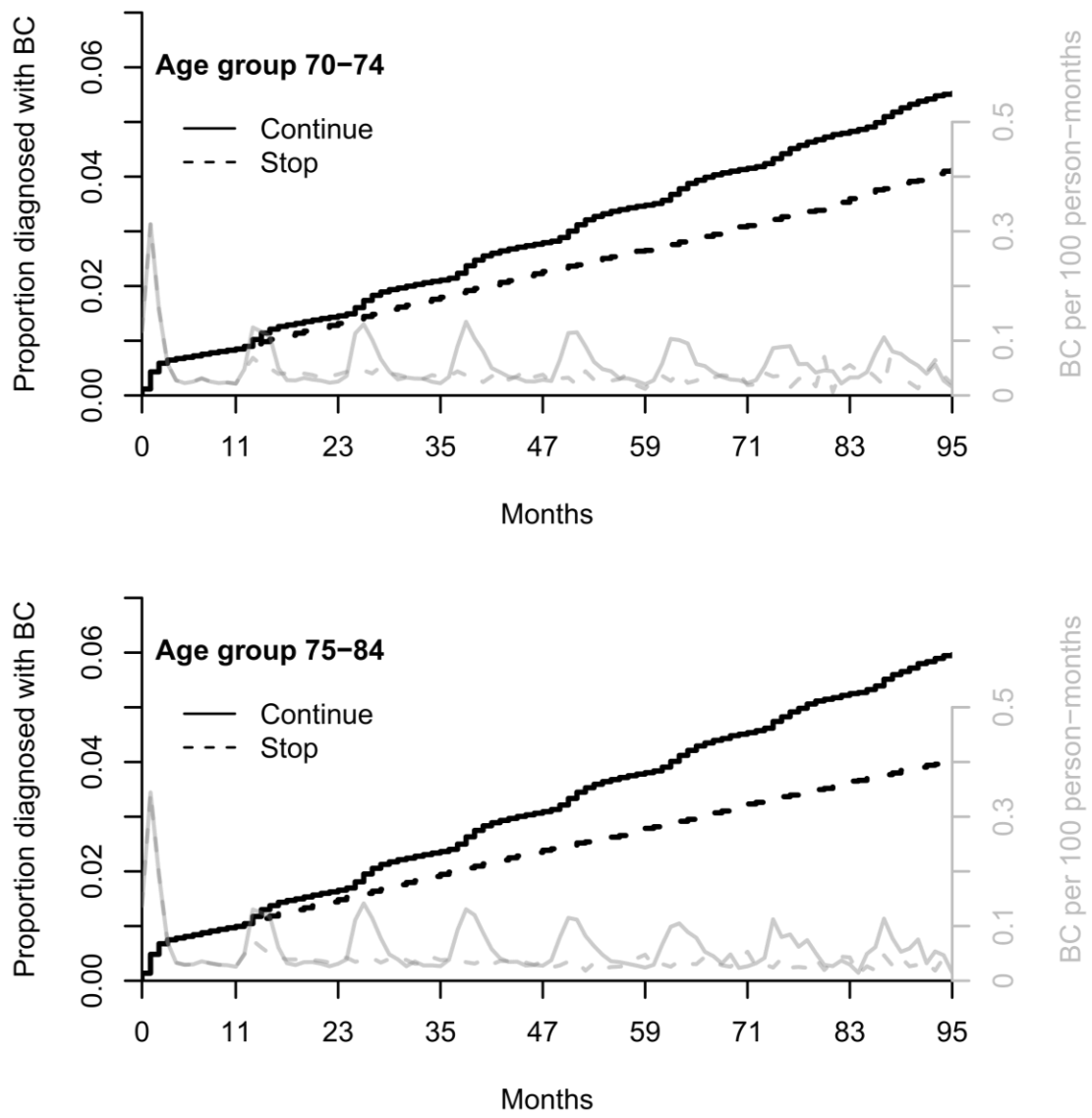
HR: hazard ratio.

* Age, calendar year, race, census region, emergency department visit in the previous 6 months, days admitted to a hospital in the previous 6 months, combined comorbidity score, cardiovascular screening in the previous year, colorectal cancer screening in the previous year, diabetes screening in the previous year, pelvic screening in the previous year, influenza shot in the previous year, bone mass measurement in the previous year, annual wellness visit, long term care resident, diagnosis of the following CCW conditions: Alzheimer's disease and related disorders, acute myocardial infarction in the previous year, atrial fibrillation, anemia, asthma, cataracts, chronic heart failure, chronic kidney disease, chronic obstructive pulmonary disease, colorectal cancer, depression, diabetes, endometrial cancer, glaucoma, hip fracture in the previous year, hypertension, hypothyroidism, ischemic heart disease, hyperlipidemia, lung cancer, osteoporosis, rheumatoid arthritis/osteoarthritis, stroke in the previous year.

** Emergency department visit in the previous 6 months, days admitted to a hospital in the previous 6 months, combined comorbidity score, cardiovascular screening in the previous year, colorectal cancer screening in the previous year, diabetes screening in the previous year, pelvic screening in the previous year, influenza shot in the previous year, bone mass measurement in the previous year, annual wellness visit, atrial fibrillation,

Alzheimer's disease, acute myocardial infarction in the last year, anemia, asthma, cataracts, chronic heart failure, chronic kidney disease, chronic obstructive pulmonary disease, colorectal cancer, depression, diabetes, endometrial cancer, glaucoma, hip fracture in the previous year, hypertension, hypothyroidism, ischemic heart disease, hyperlipidemia, lung cancer, osteoporosis, rheumatoid arthritis/osteoarthritis, stroke in the previous year, institutionalization in a long term care center, breast symptoms in the last 6 months, number of screening mammograms received, number of diagnostic mammograms received.

Supplement Figure 4. **Breast Cancer Incidence by Screening Strategy and Age Group, Medicare 1999-2008.**



Unmeasured Confounding

A key question in causal observational analyses is whether any differences between the group-specific estimates can be exclusively attributed to the treatment strategies received by each group. For example, after adjustment for multiple factors, we found a slightly lower risk of breast cancer death under continuation of screening than under stopping of screening in women aged 70-74. A reasonable concern is whether this small difference in breast cancer death is due to unadjusted differences in the distribution of risk factors (confounders) of women who happened to follow each treatment strategy. Though the possibility of residual or unmeasured confounding can never be eliminated in an observational analysis, several steps can be followed to alleviate the concern.

First, as many potential confounders as possible need to be identified and adjusted for. In our analysis, we adjusted for a large number of baseline and time-varying variables which could be associated with both screening and breast cancer death, including baseline variables age, calendar year, race, census region, visits to the ER in the 6 months before baseline, days admitted in the 6 months before baseline, combined comorbidity score at baseline; reception of the following in the year before baseline: influenza vaccine, bone mass measurement, CVD screening, diabetes screening, annual wellness visit, colorectal cancer screening, pelvic screening; diagnosis of the following before baseline: Alzheimer's disease and related disorders, acute myocardial infarction in the previous year, atrial fibrillation, anemia, asthma, cataracts, chronic heart failure, chronic kidney disease, COPD, colorectal cancer, depression, diabetes, endometrial cancer, glaucoma, hip fracture in the previous year, hypertension, hypothyroidism, ischemic heart disease, hyperlipidemia, lung cancer, osteoporosis, rheumatoid arthritis/osteoarthritis, stroke in the previous year; and time-varying variables time since last mammogram; type (screening, diagnostic) and number of mammograms received; acute myocardial infarction, hip fracture, stroke, and use of preventive services in the previous 12 months; breast symptoms, visits to the emergency department, and days admitted in the previous 6 months; combined comorbidity score; institutionalization in a long-term care facility; new diagnosis of any of the Medicare Chronic Conditions Data Warehouse comorbidities in Table 2.

Second, measuring all important confounders is not sufficient: g-methods need to be used to adjust for time-varying confounders that may be affected (or share causes with) prior treatment (1). In our analysis, we used inverse probability weighting to appropriately adjust for time-varying confounders.

Third, negative outcome controls can sometimes be used to provide guidance about the expected magnitude of the unmeasured confounding (2). Some unmeasured variables that may be associated with breast cancer screening and breast cancer mortality are history of familial gynecologic cancer, obesity and use of postmenopausal hormone therapy. These factors may also be associated with mortality from cancer of the uterus, but breast cancer screening does not cause or prevent death from cancer of the uterus. Therefore, we used death from cancer of the corpus uteri (endometrium, myometrium, fundus and isthmus uteri, but not cervix because the main risk factor for cervical cancer is human papillomavirus infection rather than family history or hormonal profile) as a negative control outcome (2). The risk differences of death from cancer of the uterus was small and the hazard ratios were similar to those for death from breast cancer (though all these estimates were imprecise). Because death from cancer of the corpus uteri is not affected by mammography screening, the apparent benefit suggests that breast cancer screening is associated with other related screening or health care. These findings suggest that our effect estimates are not underestimating the benefit of cancer screening on breast cancer death.

Fourth, the impact of adjustment for measured confounders need to be evaluated. In our analyses, unadjusted effect estimates suggested a beneficial effect of screening in all age groups, but adjustment for the measured confounders moved the effect estimates close to the null for women aged 70-74 and essentially to the null for

older women. Since the adjusted confounders are expected to be imperfectly measured in any healthcare database, one can reasonably assume that the estimates in the younger group might have moved even closer to the null had the confounders been perfectly measured. Again, our estimates may be exaggerating the benefit of cancer screening on breast cancer death.

Fifth, a sensitivity analysis for unmeasured confounding may be conducted. However, this approach has serious conceptual and technical limits, which we now describe.

A sensitivity analysis for unmeasured confounding can only conclude that, if there is much unmeasured confounding, there will be much bias in the estimates and, conversely, if there is little unmeasured confounding, there will be little bias in the estimates. But, if the magnitude of unmeasured confounding is unknown (as it often is), then the magnitude of confounding bias is also unknown. That is, sensitivity analysis for unmeasured confounding is, by definition, uninformative. The situation is different when investigators have some information about the unmeasured confounder(s). For example, our analysis did not adjust for cigarette smoking, which may be associated with both breast cancer screening (because smokers are less likely to use preventive services) and breast cancer death. If we knew the prevalence of smoking in screening-defined groups and the association between smoking and breast cancer death, then we could use published methods to correct the confounded effect estimates.

Unfortunately, these methods assume that the residual confounding can be summarized by a single variable that happens to be dichotomous, time-fixed, and unassociated with all measured variables that were adjusted for in the analysis (3,4). Supplement Tables 3 and 4 are an example of the application of those methods. Supplement Table 3 shows the corrected effect estimates under several hypothetical scenarios defined by the prevalence of the confounder in each treatment group and its association with the outcome. Supplement Table 4 shows the results of a probabilistic bias analysis that yields the distribution of the effect estimates when, rather than defining fixed scenarios, we assumed a statistical distribution for the unmeasured confounder. Specifically, we assumed that the prevalence of the unmeasured confounder in the “continue screening” arm had an equal probability density between the values 2.5% and 10% and a linearly decreasing probability to the minimum and maximum values of 0.5% and 25% (trapezoidal distribution). We also assumed a trapezoidal distribution of the unmeasured confounder in the “stop screening” arm with a minimum of 0.1%, a maximum of 20% and modes of 1% and 5%. Similarly, we assumed a trapezoidal distribution of the 8-year breast cancer death risk difference in cases per 1000 individuals between levels of the unmeasured confounder with a minimum of -1, a maximum of 0, and modes of -0.5 and -0.25. Estimates are based on 100,000 random samples from those distributions. All these analyses suggest that our effect estimates would not change much under the scenarios we considered.

Had we chosen more extreme distributions of the bias parameters, the differences would be bigger, i.e. in the presence of a weak, unmeasured, baseline confounder, our estimates would be similar, and in the presence of a strong, unmeasured confounder, our estimates would be different. Again, because we have no information about the unmeasured confounder, we cannot logically conclude whether our estimates are little biased or very biased.

These forms of sensitivity analyses for unmeasured confounding make extreme simplifications. In many settings, unmeasured confounding is probably due to the existence of multiple time-fixed and time-varying, discrete and continuous variables that are associated with the adjustment variables. It is possible that none of these variables in isolation is sufficient to impact the effect estimate but that, as a set of variables, they introduce substantial confounding. Attempting to summarize this complex vector of variables into a single dichotomous, time-fixed unmeasured variable unassociated with the measured variables may be impossible. In analyses in which, like ours, all individuals are included in both groups at baseline, individuals in both groups

have the same distribution of baseline confounders by definition. Therefore, sensitivity analyses for unmeasured confounding would need to focus on post-baseline (time-varying) unmeasured factors.

In summary, concerns about unmeasured confounding in observational analyses cannot be eliminated but can be allayed by appropriate adjustment for multiple time-fixed and time-varying prognostic factors, as well as the use of negative controls and the comparisons of adjusted and unadjusted estimates. Sensitivity analyses for unmeasured confounding are not generally informative in the absence of detailed information about the unmeasured confounders.

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